

Report for exercise BE 01 from group 01

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AI Disclaimer

OpenAI ChatGPT 5.5 was used in the creation of this report. Specifically:

- discussing code results
- latex formulas
- latex tables
- debugging numpy issues
- creating code for plots

Report on task 1 Sobol Global Sensitivity Analysis

Global Sensitivity Analysis investigates a system (e.g. the damped linear oscillator) in terms of its sensitivity with respect to the uncertain input parameters $(c, k, f, y_0$ and $y_1)$. It answers the question how susceptible the dependent variable $y(10)$ is changes in the parameters.

To apply Global Sensitivity Analysis this task makes use of first-order and total Sobol indices that are calculated through (1) a monte-carlo approximation according to Eq.1 and (2) a PCE approximation.

The monte-carlo approximation can be calculated in a number of ways but here the following is chosen:

$$\begin{aligned} V_{X_i}(\mathbb{E}_{\mathbf{X}_{\sim i}}(Y|X_i)) &\approx \frac{1}{N} \sum_{j=1}^N f(\mathbf{B})_j \left(f(\mathbf{A}_{\mathbf{B}}^{(i)})_j - f(\mathbf{A})_j \right) \\ \mathbb{E}_{\mathbf{X}_{\sim i}}(V_{X_i}(Y|\mathbf{X}_{\sim i})) &\approx \frac{1}{2N} \sum_{j=1}^N \left(f(\mathbf{A})_j - f(\mathbf{A}_{\mathbf{B}}^{(i)})_j \right)^2 \end{aligned} \quad (1)$$

, where $\mathbf{X}$ is the vector of uncertain quantities, (c, k, f, y_0, y_1) in our case, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{N \times D}$ are sample matrices of the $D = 5$ uncertain quantities and $f(\cdot)$ is the damped linear oscillator. This and other approximations are compared in [1], where also the notation is further explained.

From these quantities it is easy to calculate the first-order and total Sobol indices as follows:

$$\begin{aligned} S_i &= \frac{V_{X_i}(\mathbb{E}_{\mathbf{X}_{\sim i}}(Y|X_i))}{V(Y)} \\ S_{T_i} &= \frac{\mathbb{E}_{\mathbf{X}_{\sim i}}(V_{X_i}(Y|\mathbf{X}_{\sim i}))}{V(Y)} \end{aligned} \quad (2)$$

, where $V(Y)$ is the (total) variance of the dependent variable Y (or in our case $y(10)$).

These approximations are then compared against a calculation of Sobol indices through a PCE approximation of the damped linear oscillator $f(\mathbf{X}, t) \approx \sum_{\mathbf{n} \in A} \hat{f}_{\mathbf{n}}(t) \phi_{\mathbf{n}}(\mathbf{X})$. The partial variances and also Sobol indices are fairly easy to obtain once the PCE approximation has been constructed. As the different orders of interactions between the input parameters, i.e. the polynomials, are orthogonal to each other, the partial variance introduced by the singular effects of input parameter X_i can be calculated through

$$V_i = \sum_{\{n \in A: n_i \neq 0 \wedge \forall i \neq j: n_j = 0\}} \hat{f}_{\mathbf{n}}^2$$

, the sum of squared coefficients over the monomials in which only X_i appears. Also the total partial variance, introduced by all interactions related to X_i can be obtained via:

$$V_{T_i} = \sum_{\{n \in A: n_i \neq 0\}} \hat{f}_{\mathbf{n}}^2.$$

A , not to be confused with the one in Eq. 1, is an index set, where a given entry $(n_1, n_2, n_3, n_4, n_5)$ indexes the monomial of the PCE approximation, where parameter X_i has exponent n_i .

The partial variances V_i and V_{T_i} are normed on the variance of Y to obtain first-order and total Sobol indices.

Implementing the monte-carlo approach with $N = 893$ (more on that choice later) and the PCE approach with any exponent being at most $N \in \{3, 4\}$ and $K \in \{3, 4\}$ as quadrature degree, yields the results seen in Fig. 1.

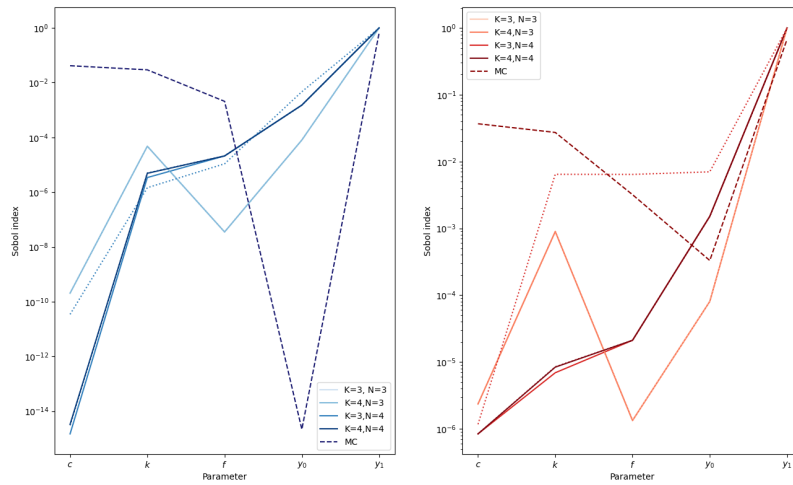


Figure 1: The Sobol indices of first-order (solid lines) and total (dotted lines) obtained through PCE approximation vs monte-carlo sampling with 893 samples.

It becomes clear that the monte-carlo sampling gives different results from the PCE approximations, especially for first-order Sobol indices. For higher sample sizes, the monte-carlo indices align better with PCE indices, indicating that PCE results are more accurate in this case. This is backed by the assumption that the model solution is smooth w.r.t. the parameters, meaning that the PCE model can better approximate the real model.

The corresponding compute times for the different approaches can be seen in Fig. 2.

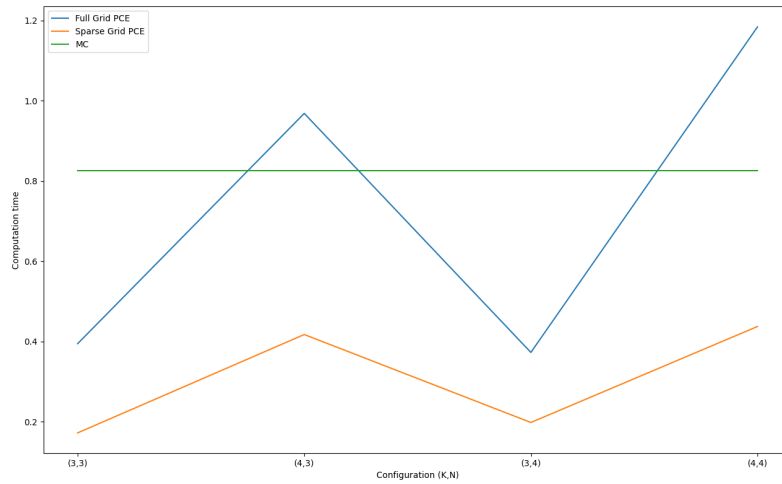


Figure 2: The compute time for different configurations of K and N compared against the monte-carlo calculation.

This result visualizes the fact that, especially full-grid PCE heavily depends on K , the number of quadrature nodes and less on the polynomial degree.

In Eq. 1 it can be observed how the calculation of Sobol indices takes $f(\mathbf{A})$, $f(\mathbf{B})$ and also $f(\mathbf{A}_B^{(i)})$ for every $i \in \{1 \dots D\}$. In total that amounts to $N \cdot (D + 2)$ solves for the approximation of both sets of indices.

On the other hand PCE approximation with a full grid of quadrature nodes gives for order K a grid of $(K+1)^D$ nodes in a D -dimensional grid. Setting the total number of combined samples in matrices $\mathbf{A}, \mathbf{B}$ of the MC approach to be $N_{MC} = 2 \cdot \frac{(K+1)^D}{D+2} \approx 893$ gives then a similar runtime.

In conclusion, the calculation of Sobol indices through monte-carlo achieves worse results than the PCE approximations with comparable compute time. Hence with respect to both qualities PCE is a better approach in this case.

Report on task 2 Gaussian Random Fields

A Gaussian random Field is defined through its mean and covariance function μ and Σ respectively. The choice of covariance function especially has a great effect on the resulting realizations / samples of the random field and how different parts of the field interact with each other.

Two popular choices for covariance functions are the exponential kernel and the squared exponential kernel, both of which are used in the following to generate samples.

To sample a field over a given domain, you first need to discretize your sample points within that domain. Then, after computing the pair-wise covariance / the kernel values $K(x, x')$ for any two x, x' in the discretized domain, samples of the Gaussian Random Field are jointly Gaussian following $\mathcal{N}(\mu_{\mathbf{x}}, \Sigma)$ where $\Sigma_{ij} = K(x_i, x_j)$, with K being the kernel of choice.

With the mean constant i.e. $\mu_{\mathbf{x}} = \mu = 0.1$ and the grid being defined over the cube $[0, 1]^2$, sampling 3 instances of a Gaussian Random Field becomes just a sample of a multivariate Gaussian. Examples can be seen in Fig. 3 and Fig. 4.

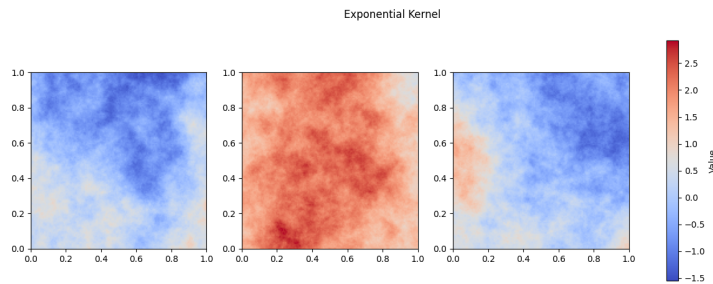


Figure 3: 3 realizations of a Gaussian Random Field over $[0, 1]^2$ with an exponential kernel function that has scale parameter $l = 2$.

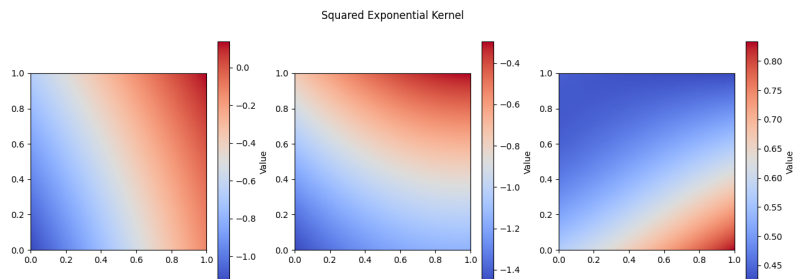


Figure 4: 3 realizations of a Gaussian Random Field over $[0, 1]^2$ with a squared exponential kernel function and scale parameter $l = 2$.

Evidently transitions between high-value regions and low-value regions are much smoother in the case of the squared exponential kernel. This follows a similar behavior as the decay of a parabola towards $x = 0$ in comparison to the absolute value function. The square in the squared exponential kernel induces a smoother decay in covariance between increasingly distant points on the grid.

When modeling some natural phenomenon with a Gaussian Random Field, the kernel function becomes a central design choice. From the before observed differences between ordinary and squared exponential kernel it can be concluded that a squared exponential kernel is more suitable for phenomena with smoothly decaying local influences, where close-by points are tightly coupled (e.g. heat transfer). The ordinary exponential kernel on the other hand might be better suited for sharp changes between close points. Such use cases might include modeling material structure with defects (fatigue/cracks in some places need not imply defects in the immediate vicinity).

Report on task 3 Wiener Process

3.1

The Wiener Process is a stochastic process in continuous time. To implement it, we discretize the time into N steps, in which the process moves for i.i.d values from the last state. So N represents a discrete time space. Total time $T = [0, 1]$ is held for this subtask, as are the number of steps in this time of $N = n_points = 1000$. This gives a $\Delta t = 1e^{-3}$. The path moves about $\mathcal{N}(0, \sqrt{\Delta t})$ from the previous state, leading to paths such as the one depicted in figure 5, generated using:

$$W_{t_{j+1}} = W_{t_j} + \sqrt{\Delta t} Z_j, \quad Z_j \sim \mathcal{N}(0, 1).$$

We can see the path is jagged with some high frequency oscillations but also it can also have randomly occurring peaks and troughs. Every generated path is visibly different and due to the generating process, there should be no underlying tendencies.

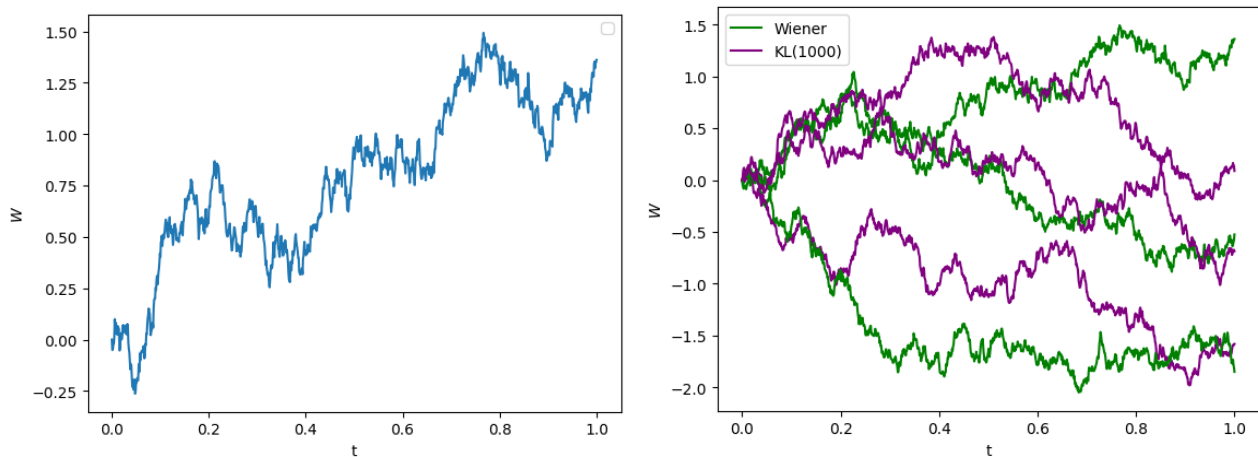


Figure 5: (left) Path of a single sample for the standard Wiener Process. (right) Multiple different paths from a Wiener Process path and a KL ($M=1000$) approximation.

Such a process can be approximated with a truncated Karhunen-Loève expansion. It sums over weighted sinusoidal eigenfunctions multiplied with gaussian noise to simulate the same sort of behavior. Truncated, since it would be an infinite-dimensional expansion, but we only select the first M modes. So M "discretizes" the KL mode space in the sense, that we truncate and use only the eigenfunctions corresponding to the larger eigenvalues.

The KL approximation is defined as follows:

$$W_t = \sum_{m=1}^{\infty} \sqrt{\lambda_m} \phi_m(t) \zeta_m \approx \sum_{m=1}^M \sqrt{\lambda_m} \phi_m(t) \zeta_m \quad \zeta_m \sim \mathcal{N}(0, 1).$$

with the eigenvalues and -functions defined as follows:

$$\lambda_m = \frac{1}{\left(m - \frac{1}{2}\right)^2 \pi^2}, \quad \phi_m(t) = \sqrt{2} \sin\left(\left(m - \frac{1}{2}\right) \pi t\right), \quad m = \text{range}(M)$$

Figure 6 (left) shows the eigenvalues generated with the above equations and $M = 1000$ nodes. It shows how the eigenvalue sizes quickly decays, and so the attributed weight to the later eigenfunctions. The eigenfunctions increase in frequency, as can be seen in the right figure, and so the earlier functions are weighed heavier, contributing more to a movement over longer time, whereas the later eigenfunctions add more high frequency movements.

This can also be clearly seen in figure 7, where paths sampled with lower values of M on the same seed only contain lower frequency oscillations. It can be said, that the path with $M = 1000$ more or less equals the directly sampled Wiener Process path in terms of noise frequency.

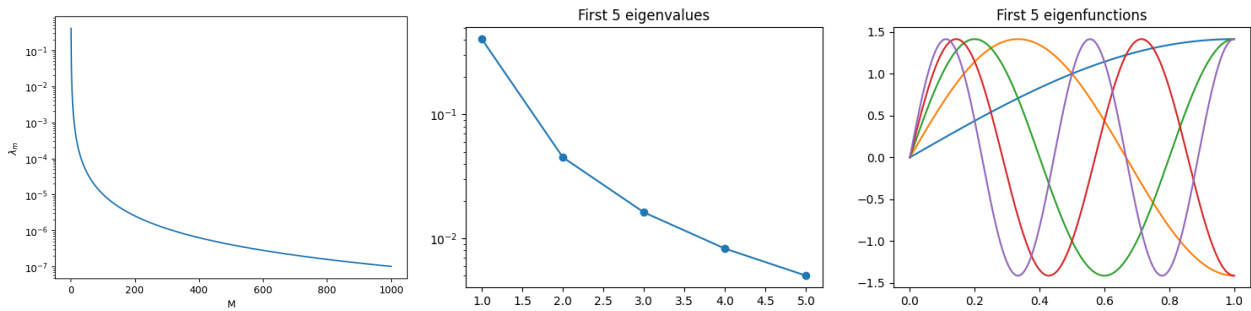


Figure 6: (left) Showing the decay of the first 1000 eigenvalues of the KL approximation. (middle) the first 5 Eigenvalues. (right) Shown are the first 5 eigenfunctions associated with the largest eigenvalues.

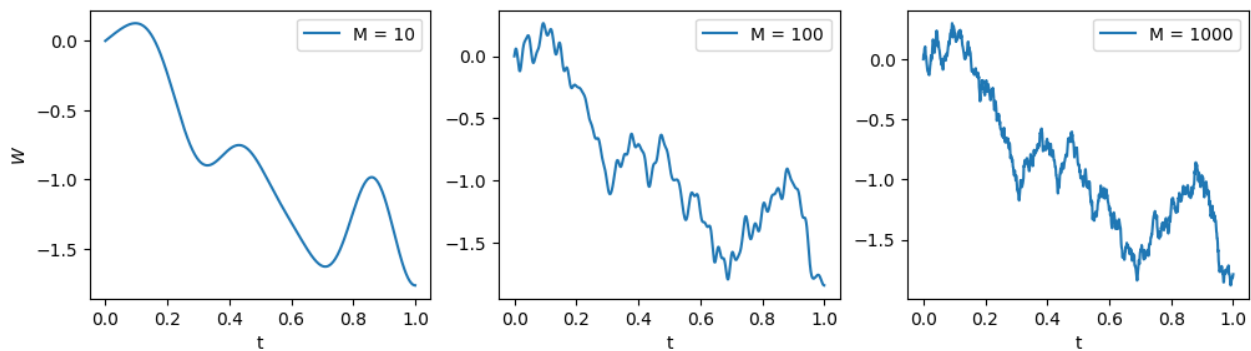


Figure 7: Differing number of modes (10, 100, 1000) for a KL approximation.

We cant set the same seed for both methods KL and Wiener to the end that the processes will be identical, as the way the standard normalized samples are used differs (direct movement vs weighting of eigenfunction modes). So we are left to qualitatively assess the similarity of the methods. In figure 5 (right), we show 3 paths sampled from both methods.

3.2

To write

change eigenfunctions and ev to accomodate for T

implementing via slicing and euler

notice that `rng.standard_normal((r, c))` are not the same numbers

describe oscillator

describe plots

section on what else could be used other than monte carlo

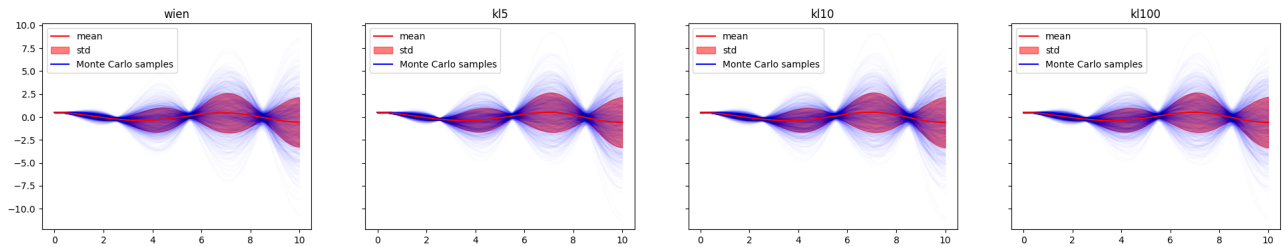


Figure 8: Paths of all 1000 sample paths evaluated by the oscillator model in blue over the different sample generating processes, see indiv figure headers. Highlighted in red are the means and std over time.

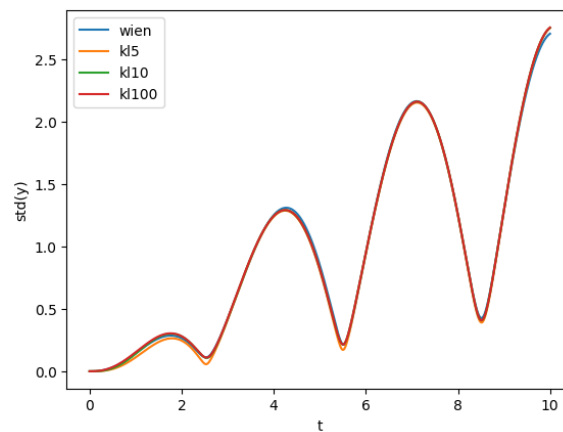


Figure 9: Standard deviation of y over 1000 samples of the oscillator model over time. Different models for creating samples highlighted in color.

	Wiener	KL-5	KL-10	KL-100
mean	-0.55653	-0.57676	-0.57940	-0.57917
var	7.32498	7.57309	7.58428	7.59484

Table 1: Metrics of the different Methods & mode numbers using monte carlo with 1000 sample paths. Evaluated at the final position of $y(10)$.

References

- [1] Andrea Saltelli, Paola Annoni, Ivano Azzini, Francesca Campolongo, Marco Ratto, and Stefano Tarantola. Variance based sensitivity analysis of model output. design and estimator for the total sensitivity index. *Computer Physics Communications*, 181(2):259–270, 2010.